Projector-Resolved Seed Calculus VII

First Lean-Checked Block

Quantum packets, circuits, symmetry and effective dynamics

Komorebi P.

Version 0.1.0 | 6 September 2026

Release result

The four active Lean files compiled successfully. They contain **32 theorem declarations**, including supporting lemmas, concrete matrix specializations and an additional constraint audit. They cover **six of the original paper's 19 named finite theorem-like items**. These are different counts. The 19-item ledger contains 15 theorems, one corollary and three classical inputs. It excludes numerical certificates, unnumbered assertions and the five continuum conjectures.

Source item	Checked mathematical content	Status
4.1	Arbitrary finite circuit path expansion and reconstruction	Pass
4.2	Both packet unitarity identities, including complex matrices	Pass
5.1	Exact symmetry for the generator and its exponential	Pass
7.1	Block-diagonal measurement invariance	Pass
9.1	Feshbach resolvent, with corrected invertibility assumptions	Pass
11.1	Second-order dispersion bound with explicit nonzero spacing	Pass

Compiler and dependency. Lean 4.34.0-rc2 was run against mathlib revision

`aeb72f1e9981b6d35b426b16dca7759be3e7ba06`.

Each source was checked with one worker and warnings treated as errors. All 32 theorem axiom reports contained only `propext`, `Classical.choice` and `Quot.sound`. There are no admitted proofs or new axiom declarations in the four active files. A false equality and a deliberate proof placeholder were both rejected with exit code 1 as expected.

What changed mathematically. The Feshbach theorem needs a full-resolvent invertibility hypothesis in addition to invertibility of the eliminated block. The constraint section also needs a sharper distinction between compressed anomaly vanishing and operator closure. Both issues are addressed below, with checked counterexamples.

Verification boundary. *Thirteen named finite items remain unchecked in this release. None of the five continuum conjectures is proved. The original numerical suite and the full Papers I–VI proof collections were not rerun. The two reused Paper VI modules were adapted and freshly compiled here. This supplement does not replace the entire original manuscript or certify every assertion in it.*

1. Exact packets and arbitrary circuit length

Let R be a unital ring, not necessarily commutative, and let I be a finite address set. An exact packet family is a family $P_a \in R$ such that

$$P_a P_b = \begin{cases} P_a, & a = b, \\ 0, & a \neq b, \end{cases} \qquad \sum_{a \in I} P_a = 1.$$

Adjoint conditions are added when needed. For an operator U, define the transfer corner

$$U_{ba} = P_b U P_a.$$

The target index comes first. Completeness and distributivity give

$$\sum_{b,a} U_{ba} = U, \qquad (UV)_{ba} = \sum_c U_{bc} V_{ca}.$$

The second identity uses $P_c^2 = P_c$ when adjacent transfer corners are multiplied.

Circuit paths: source Theorem 4.1

For $U = U_m \cdots U_1$, the checked result is

$$P_{a_m} U P_{a_0} = \sum_{a_1, \ldots, a_{m-1}} P_{a_m} U_m P_{a_{m-1}} \cdots P_{a_1} U_1 P_{a_0}.$$

The Lean implementation defines a recursive path sum and proves the identity by induction over an arbitrary finite gate list. It is not restricted to a fixed circuit depth. The empty list has circuit operator 1 and path value $P_b P_a$. Summing the source and target indices reconstructs the full circuit.

Formal hooks. `PacketFamily.compose_blocks`, `PacketFamily.circuit_path_expansion` and `PacketFamily.circuit_reconstruction` in `QuantumPacketCore.lean`. The gate list is ordered from latest to earliest, $[U_m, \ldots, U_1]$.

Adjoint conservation: source Theorem 4.2

Assume a star operation with $P_a^* = P_a$. Then

$$(U_{ba})^* = P_a U^* P_b.$$

The two checked conservation laws are

$$U^*U = 1 \quad \Longrightarrow \quad \sum_b (U_{ba})^* U_{bc} = \delta_{ac} P_a,$$

$$UU^* = 1 \quad \Longrightarrow \quad \sum_a U_{ba} (U_{da})^* = \delta_{bd} P_b.$$

For the first law, expand each corner, collapse P_b^2 and sum:

$$\sum_b P_a U^* P_b U P_c = P_a U^* \left(\sum_b P_b \right) U P_c = P_a P_c.$$

The second law is analogous and is separately checked. Concrete complex-matrix versions use conjugate transpose as the adjoint. No commutative multiplication rule is used for operator products.

Formal hooks. `unitarity_columns`, `unitarity_rows`, `matrix_unitarity_columns` and `matrix_unitarity_rows`.

2. Symmetry and measurement

Exact symmetry: source Theorem 5.1

If $HP_a = P_aH$ for every packet, then

$$P_bHP_a = 0 \quad (b \neq a), \qquad H = \sum_a P_aHP_a.$$

In a complex Banach algebra, commutation also propagates to the actual exponential:

$$e^{zH}P_a = P_ae^{zH} \qquad (z \in \mathbb{C}).$$

Consequently,

$$e^{zH} = \sum_a P_ae^{zH}P_a, \qquad P_be^{zH}P_a = 0 \quad (b \neq a).$$

This covers real-time evolution by taking $z = -it$. The Lean proof invokes the exponential commutation theorem, not an abstract flow supplied with the desired invariant-packet property as a hypothesis. Self-adjointness of H is not needed for the algebraic reducing property, so this result applies more broadly than the Hermitian specialization in the paper.

Formal hooks. `PacketFamily.symmetry_diagonal`, `PacketFamily.symmetry_exponential_commutes`, `PacketFamily.symmetry_exponential_diagonal` and `PacketFamily.symmetry_exponential_no_transfer`.

This is an exact symmetry result. It does not supply the quantitative broken-symmetry leakage estimate of Theorem 5.2, whose Duhamel and norm arguments remain outside this block.

Measurements: source Theorem 7.1

For finite complex matrices, define packet dephasing by

$$\mathcal{D}_P(X) = \sum_a P_aXP_a.$$

The code proves idempotence and trace duality:

$$\mathcal{D}_P(\mathcal{D}_P(X)) = \mathcal{D}_P(X), \qquad \mathrm{Tr}(\mathcal{D}_P(E)\rho) = \mathrm{Tr}(E\mathcal{D}_P(\rho)).$$

Trace duality follows term by term from cyclicity:

$$\mathrm{Tr}(P_aEP_a\rho) = \mathrm{Tr}(EP_a\rho P_a).$$

Therefore a packet-diagonal effect, $\mathcal{D}_P(E) = E$, satisfies

$$\boxed{\mathrm{Tr}(E\rho) = \mathrm{Tr}(E\mathcal{D}_P(\rho)).}$$

The proof holds for arbitrary finite complex matrices E and ρ. Density matrices and positive measurement effects are valid specializations rather than assumptions needed for this trace identity. Taking $E = 1$ also gives trace preservation of packet dephasing.

Formal hooks. `dephase_trace_duality`, `measurement_invariance` and `dephase_trace_preserving`.

This does not establish the full Kraus-channel theorem or the Knill–Laflamme recovery criterion. Those require additional positivity, auxiliary-system or recovery-map arguments.

3. Corrected Feshbach resolvent

Write a finite complex matrix in retained and eliminated coordinates:

$$H = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad zI - H = \begin{pmatrix} zI_P - A & -B \\ -C & zI_Q - D \end{pmatrix}.$$

Set

$$D_z = zI_Q - D, \qquad S_z = zI_P - A - BD_z^{-1}C.$$

The effective Hamiltonian is $H_{\text{eff}}(z) = A + BD_z^{-1}C$, so $S_z = zI_P - H_{\text{eff}}(z)$.

Correct statement: source Theorem 9.1

Assume both D_z and $zI - H$ are invertible. Then S_z is invertible and

$$\boxed{\left((zI - H)^{-1}\right)_{PP} = S_z^{-1}.}$$

The left-hand side is the literal retained-coordinate submatrix of the full inverse. This matches the paper's compressed operator when written in an orthonormal basis of the retained subspace. The proof is valid for arbitrary complex blocks of compatible sizes and does not require Hermiticity.

The checked invertibility criterion is

$$D_z \text{ invertible} \quad \Longrightarrow \quad \big(zI - H \text{ invertible} \iff S_z \text{ invertible}\big).$$

Thus one may equivalently assume invertibility of D_z and S_z.

Why the original assumption is insufficient. Take $z = 0$ and $H = \operatorname{diag}(0, -1)$, retaining the first coordinate. Then

$$D_z = 1, \qquad zI - H = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

The eliminated block is invertible, but the full matrix is singular. Hence the original hypothesis does not ensure the existence of its stated resolvent. The Lean counterexample proves invertibility of the scalar 1 and noninvertibility of this full 2×2 matrix.

Formal hooks. In `SchurFeshbach.lean`: `projected_resolvent`, `schur_invertible` and `eliminated_invertible_full_singular`. The formal invertibility assumptions are genuine `Invertible` instances, not assumptions of the desired resolvent identity.

Provenance and limits

The general block-inverse proof was reused from the checked fifth block of Paper VI and freshly compiled in the Paper VII namespace. The extra counterexample was added in this release. This is reuse of a proof, not a claim of a new Schur identity.

The unnumbered time-domain memory equation in Section 9.2, cutoff-uniform resolvent control and continuum limits are not proved by this finite inverse identity. The source amendment makes the two invertibility assumptions explicit without changing the intended effective Hamiltonian.

4. Dispersion with an explicit spacing condition

The grid eigenvalue in source Theorem 11.1 is

$$\lambda_{k,h} = \frac{4}{h^2} \sin^2\left(\frac{kh}{2}\right).$$

The original low-mode condition is $0 \le kh/2 \le 1$. A nonzero spacing is also required for the quotient to have its intended meaning. For a physical grid, one normally takes $h > 0$.

Checked stronger statement

For every $k, h \in \mathbb{R}$ with $h \neq 0$,

$$\boxed{0 \le k^2 - \lambda_{k,h} \le \frac{k^4 h^2}{12}.}$$

The code proves this without the low-mode restriction. The bound may be loose away from the low-mode regime, but it remains valid.

The proof begins with the all-real estimate

$$0 \le x^2 - \sin^2 x \le \frac{x^4}{3}.$$

The lower bound follows from $|\sin x| \le |x|$. For the upper bound, use

$$|x - \sin x| \le \frac{|x|^3}{6}, \qquad |x + \sin x| \le 2|x|.$$

Factoring the difference of squares gives

$$0 \le x^2 - \sin^2 x = |x - \sin x|\,|x + \sin x| \le \frac{x^4}{3}.$$

With $x = kh/2$ and $h \neq 0$,

$$k^2 - \lambda_{k,h} = \frac{4}{h^2}\left(x^2 - \sin^2 x\right),$$
$$\frac{4}{h^2}\frac{x^4}{3} = \frac{k^4 h^2}{12}.$$

Every algebraic transformation and inequality in the active source was accepted by Lean.

Formal hooks. `LowModeConvergence.sine_square_error` and `LowModeConvergence.low_mode_error`. The proof body is adapted from the Paper VI low-mode module, with provenance retained.

What this does not establish

This is an exact deterministic error bound for the displayed dispersion relation. It is not a verification of the original paper's floating-point convergence-rate experiment, spectral-dimension plateau measurements or selection of a physical spacetime geometry. Those computations were not rerun in this release.

5. A constraint diagnostic that vanishes automatically

Section 12 defines the physical projector using the exact common kernel of the constraints. In finite dimensions this gives $C_a P = 0$ for every a. Absorb the factor i and any structure coefficient into f_c, and write

$$\mathcal{A}_{ab} = C_a C_b - C_b C_a - \sum_c f_c C_c.$$

Proposition 1 (Exact common-kernel compression)**.** *If $C_c P = 0$ for every c, then*

$$\mathcal{A}_{ab} P = 0, \qquad P \mathcal{A}_{ab} P = 0.$$

No idempotence or adjoint property of P is needed for this implication.

Proof. Associativity gives

$$\mathcal{A}_{ab} P = C_a (C_b P) - C_b (C_a P) - \sum_c f_c (C_c P) = 0.$$

Left multiplication by P gives the second identity. □

Compressed vanishing without operator closure

Over the rationals, take

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

These displayed matrices are symmetric. The code checks that $P^2 = P$, $P \neq 0$ and $XP = ZP = 0$, but

$$[X, Z] = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -2 \\ 0 & 2 & 0 \end{pmatrix} \neq 0, \qquad P[X, Z]P = 0.$$

The commutator therefore violates the zero-structure-coefficient closure relation outside the physical range, despite exact vanishing of its physical compression.

Formal hooks. `ConstraintAudit.anomaly_annihilates_common_kernel`, `compressed_anomaly_zero`, `commutator_nonzero` and `physical_commutator_zero`, together with the nonzero-projector and annihilation checks.

Interpretation for the paper

Compressed anomaly vanishing is not sufficient to certify full operator closure. In the exact finite common-kernel setup it is automatic. A nonzero computed compression would instead contradict those exact assumptions or expose an approximation or implementation defect.

This does not establish that a nontrivial compatible refinement tower exists. It does not prove or disprove the full continuum program. A stronger closure claim must test more than this compression, and unbounded constraints need their operator domains specified. This audit is additional to the six checked source items and is not counted as a seventh original theorem.

6. Remaining proof obligations and reproduction

Item	Named source assertion	This release
2.1	Orbit dimension of one quantum seed	Not checked
2.2	Quantum seed-rank formula	Not checked
2.3	Local orbit and Schmidt rank	Not checked
2.4	Computational-basis seed	Not checked
3.1	Cyclic quantum projectors	Not checked
3.2	Tensor/polyphase refinement	Not checked
4.1	Exact circuit path expansion	Checked
4.2	Packetwise unitarity conservation	Checked
5.1	Exact symmetry packets	Checked
5.2	Commutator leakage bound	Not checked
6.1	Moving-frame Hamiltonian	Not checked
6.3	Canonical packet connection	Not checked
6.4	Projected connection, classical input	Not checked
7.1	Block-diagonal measurement invariance	Checked
7.2	Finite Kraus channel, classical input	Not checked
8.1	Knill–Laflamme criterion, classical input	Not checked
9.1	Feshbach resolvent with corrected hypotheses	Checked
10.1	Finite Euclidean transfer	Not checked
11.1	Low-mode second-order bound	Checked

The five conjectures in Section 12 remain separate: compatible anomaly-free refinement, uniform Feshbach control, Lorentzian reconstruction, an Einstein infrared limit and a physical massless spin-two sector. Unnumbered assertions and the old numerical certificates also require separate checks.

Reproduction. From the extracted release directory, with the portable Linux kit available, run

```
python3 scripts/verify.py --kit /absolute/path/to/portable-kit
```

The script checks the compiler version and actual mathlib revision, then compiles all four active files, audits each theorem's axiom dependencies and runs the expected-failure controls. It writes new evidence to a timestamped directory and exits nonzero if a gate fails. The released results remain in `verification/release_run/`.

Recorded runtime. The four final module checks took 15.654 seconds combined in the warmed session, including launcher setup. The runner releases Lake before starting Lean, uses one worker and relies on targeted module imports. No peak-memory reduction was benchmarked. The large portable runtime is not duplicated in this proof bundle.

Failure record. Early drafts failed on Lean elaboration and finite-index simplification, and one import was deprecated. The errors were repaired without admitting results. Development logs are retained, and the final run is warning-free. Imported mathlib modules were used from the pinned cache, not independently reverified in full.

The unchanged original PDF, the precise source amendment patch, the active Lean sources, the upstream proof copies and a machine-readable coverage ledger accompany this supplement.